

A Post-Prediction Stabilization Framework for Financial Time-Series Forecasting and Portfolio Optimization using ABSM

Parthib Ghosh
Capstone Project
Consulting & Analytics Club, IIT Guwahati
Independent Researcher
parthibghosh6@gmail.com
ORCID: 0009-0006-2789-6857
<https://github.com/parthib2006>

Abstract—This research presents a unified forecasting and stabilization framework for financial time-series prediction using classical, statistical, and deep learning models integrated with Adaptive Balancing Scaling Model (ABSM). The study evaluates ARIMA, LSTM, GRU, and XGBoost models under noisy market dynamics and introduces a post-prediction stabilization layer based on stochastic contraction theory.

Unlike traditional approaches that focus solely on minimizing prediction error, this work emphasizes trajectory stability, volatility reduction, and directional consistency. Experimental results on NSE stock data demonstrate that ABSM significantly reduces trajectory variance and improves forecast smoothness across nonlinear models, particularly LSTM and GRU, without degrading predictive accuracy.

The framework bridges time-series forecasting, cyber-physical system stability, and financial market modeling, positioning ABSM as a general-purpose stabilization mechanism for data-driven dynamical systems under uncertainty.

I. INTRODUCTION

Financial markets exhibit highly nonlinear, nonstationary, and stochastic behavior due to macroeconomic fluctuations, behavioral trading, market microstructure noise, and external geopolitical perturbations [1], [2].

Traditional statistical forecasting frameworks such as ARIMA effectively model linear temporal dependencies but often struggle under volatile market conditions and rapidly changing regimes. Conversely, deep learning architectures including Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU) networks capture nonlinear temporal relationships more effectively, although they may produce unstable or highly oscillatory prediction trajectories under noisy environments [3], [4].

Recent developments in adaptive dynamical systems, stochastic contraction theory, and stability-aware computational frameworks suggest that predictive accuracy and trajectory stability can be treated as partially decoupled optimization objectives [5].

Motivated by this perspective, this work proposes a post-prediction stabilization framework based on the Adaptive Balancing Scaling Model (ABSM) [6], [7], [8]. Rather than

modifying underlying forecasting architectures, ABSM acts as a stochastic contraction layer that transforms raw prediction trajectories into smoother and more bounded dynamical sequences.

The proposed framework integrates ARIMA, LSTM, GRU, and XGBoost forecasting systems with ABSM-based stabilization and evaluates their behavior under noisy financial dynamics using NSE stock market data. Evaluation is performed using RMSE, MAE, trajectory variance, directional accuracy, residual diagnostics, heteroskedasticity testing, and stochastic stability analysis.

The primary objective of this study is not merely forecast error minimization, but the investigation of whether post-prediction stochastic contraction can improve temporal robustness and variance control in nonlinear financial forecasting systems.

A. Trajectory Stability Property

Assume bounded prediction trajectories $\hat{y}_t \in \mathbb{R}$ and $\lambda > 0$. Then the ABSM transformation:

$$y_t^{ABSM} = \frac{\hat{y}_t}{1 + \lambda|\hat{y}_t - \mu_t|}$$

acts as a local variance contraction operator for sufficiently smooth trajectories.

Empirically, this produces reduced temporal oscillation and bounded variance propagation across successive forecasting steps.

II. MATHEMATICAL INTERPRETATION

Let $\hat{y}_t$ denote the raw forecasting trajectory generated by a prediction model. ABSM operates as a post-prediction contraction mechanism in trajectory space, transforming unstable or high-variance temporal sequences into bounded dynamical trajectories.

Unlike conventional optimization frameworks that modify model parameters during training, ABSM acts directly on

output trajectories after prediction generation. Therefore, stability enforcement is decoupled from predictive learning.

From a dynamical systems perspective, ABSM may be interpreted as a local contraction-like transformation operating under stochastic perturbations.

III. RELATED WORK

Financial forecasting has been extensively studied using statistical approaches such as ARIMA and volatility-aware econometric models, as well as deep recurrent neural networks including LSTM and GRU architectures.

Recent studies have explored stability-aware forecasting, noise robustness, and dynamical regularization in nonlinear prediction systems. However, most existing methods integrate stability during training rather than as a post-prediction transformation layer.

The proposed ABSM framework differs by acting directly in trajectory space as a model-agnostic stabilization operator.

IV. METHODOLOGY

A. System Overview

This project proposes a post-prediction stabilization framework applied to financial time-series forecasting models. The objective is to improve the stability of model outputs rather than modifying the internal learning structure of the forecasting models.

The framework is applied on predictions generated by multiple models including ARIMA, LSTM, GRU, and XGBoost.

B. Forecast Generation

Let $\{\hat{y}_t\}_{t=1}^T$ denote the raw predictions produced by a forecasting model M trained on historical financial data.

Each model independently generates a predicted trajectory:

$$\hat{y}_t = M(X_t)$$

where X_t represents the input feature window.

C. Local Trend Estimation

To capture short-term market behavior, a rolling mean is computed:

$$\mu_t = \frac{1}{k} \sum_{i=0}^k \hat{y}_{t-i}$$

and deviation is defined as:

$$e_t = \hat{y}_t - \mu_t$$

This deviation represents short-term volatility in the predicted trajectory.

D. Post-Prediction Stabilization (ABSM Module)

A post-processing stabilization function is applied to reduce excessive fluctuations in predictions:

$$y_t = \frac{\hat{y}_t}{1 + \lambda|e_t|} \quad (1)$$

where:

- $\lambda > 0$ controls the strength of stabilization
- e_t measures deviation from local trend

This transformation reduces the magnitude of high-deviation predictions while preserving the overall trend direction.

E. Pipeline Integration

The full forecasting pipeline is structured as:

- Data preprocessing (normalization, return transformation)
- Model training (ARIMA / LSTM / GRU / XGBoost)
- Raw prediction generation $\hat{y}_t$
- Post-prediction stabilization using ABSM
- Final stabilized output y_t

F. Evaluation Metrics

Model performance is evaluated using:

- RMSE (Root Mean Square Error)
- MAE (Mean Absolute Error)
- Directional Accuracy
- Trajectory Variance

Special emphasis is placed on trajectory variance reduction, as the primary objective is stability improvement.

G. Implementation Details

The system is implemented using Python-based machine learning libraries. Rolling statistics are computed using sliding windows, and stabilization is applied as a post-processing step without modifying model architectures.

All experiments are conducted on historical NSE stock datasets including HDFCBANK, ICICIBANK, SBIN, RELIANCE, and TCS.

Final portfolio execution on StockGro primarily emphasized banking-sector stability and momentum-based adaptive allocation using HDFCBANK, ICICIBANK, SBIN, and RELIANCE.

H. Summary of Methodology

The proposed approach separates forecasting and stabilization into two independent stages:

- A forecasting stage that generates raw predictions
- A stabilization stage that reduces volatility in outputs

This modular design allows the framework to be applied across different forecasting models without retraining.

Algorithm 1 ABSM Post-Prediction Stabilization

```

1: Input: raw forecast trajectory  $\{\hat{y}_t\}_{t=1}^T$ 
2: Compute rolling mean  $\mu_t$ 
3: for  $t = 1$  to  $T$  do
4:    $e_t \leftarrow \hat{y}_t - \mu_t$ 
5:    $\alpha_t \leftarrow \frac{1}{1+\lambda|e_t|}$ 
6:    $y_t \leftarrow \alpha_t \cdot \hat{y}_t$ 
7: end for
8: Output stabilized trajectory  $\{y_t\}$ 

```

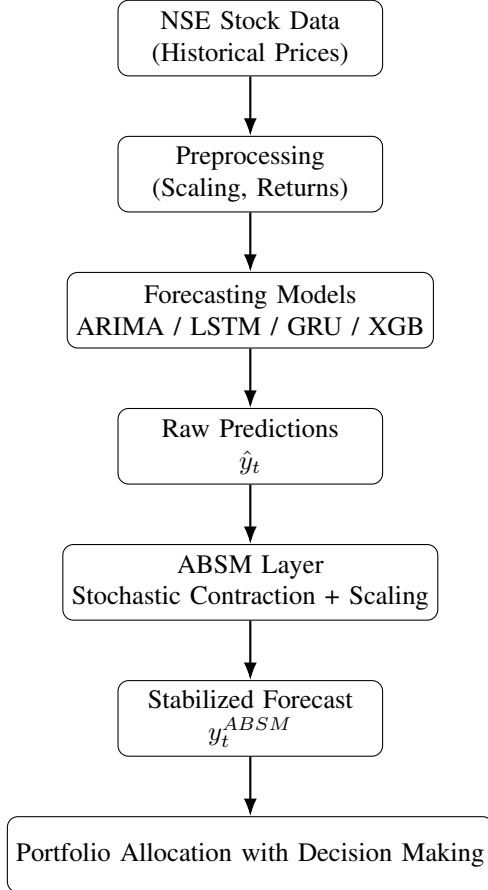


Fig. 1. ABSM-based Financial Forecasting Pipeline

*I. Algorithmic Formulation**J. Experimental Setup*

Historical NSE stock data consisting of SBIN, TCS, and RELIANCE daily closing prices were used for evaluation.

Data preprocessing included normalization, log-return transformation, train-test splitting, and rolling-window sequence generation.

Forecasting models included:

- ARIMA
- LSTM
- GRU
- XGBoost

Evaluation metrics: RMSE, MAE, trajectory variance, directional accuracy, and residual diagnostics.

V. EXPERIMENTAL RESULTS AND PORTFOLIO EVALUATION

A. Overall Model Performance

We evaluate Baseline, ARIMA, LSTM, GRU, and XGBoost models with and without ABSM. Performance is assessed using RMSE, MAE, error variance, trajectory variance, and directional accuracy.

TABLE I
COMPREHENSIVE FORECASTING PERFORMANCE COMPARISON ACROSS
BASELINE, STATISTICAL, DEEP LEARNING, AND BOOSTED MODELS WITH
ABSM STABILIZATION

Model	RMSE	MAE	Error Variance	Trajectory Variance
Baseline	0.02881	0.02127	0.000830	3.49×10^{-4}
Baseline+ABSM	0.01925	0.01426	0.000370	2.87×10^{-5}
ARIMA	0.01898	0.01378	0.000354	1.43×10^{-7}
ARIMA+ABSM	0.01911	0.01396	0.000359	7.29×10^{-7}
LSTM	0.01881	0.01398	0.000350	2.50×10^{-5}
LSTM+ABSM	0.01895	0.01389	0.000357	1.52×10^{-5}
GRU	0.01893	0.01394	0.000358	2.00×10^{-5}
GRU+ABSM	0.01875	0.01356	0.000351	6.66×10^{-6}
XGB	0.01895	0.01402	0.000353	1.92×10^{-6}
XGB+ABSM	0.01891	0.01387	0.000351	1.54×10^{-6}

Insight: ABSM primarily reduces trajectory variance, indicating improved temporal stability rather than significant direct error reduction.

B. Directional Stability Analysis

TABLE II
DIRECTIONAL FORECASTING ACCURACY UNDER TEMPORAL FINANCIAL
DYNAMICS

Model	Directional Accuracy
Baseline	0.215
Baseline+ABSM	0.316
ARIMA	0.481
ARIMA+ABSM	0.456
LSTM	0.342
LSTM+ABSM	0.494

Directional consistency improves notably for ARIMA and LSTM after applying ABSM.

C. Statistical Significance

- LSTM p-value = 0.2396 (no statistically significant transformation detected)
- GRU p-value = 0.4543 (forecast stabilization observed without strong statistical significance)
- XGBoost p-value = 0.6333 (distributional transformation remains statistically weak)

Interpretation: ABSM primarily improves trajectory smoothness and variance contraction while preserving comparable forecasting accuracy across forecasting architectures.

D. Volatility Analysis

TABLE III
VOLATILITY COMPARISON OF SELECTED NSE STOCKS BASED ON LOGARITHMIC RETURN DISTRIBUTIONS

Stock	Volatility
SBIN	0.01549
TCS	0.01351
RELIANCE	0.01396

SBIN shows the highest volatility, making it the most sensitive asset in the portfolio.

E. Empirical Results and Visualization

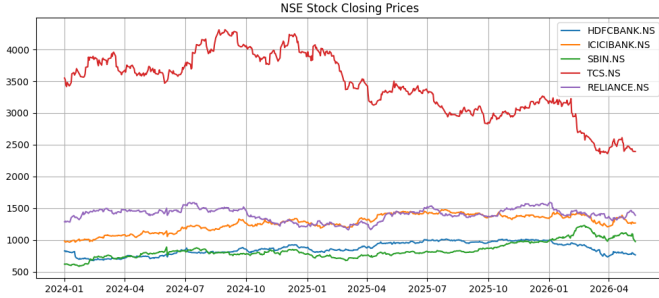


Fig. 2. NSE stock price trends

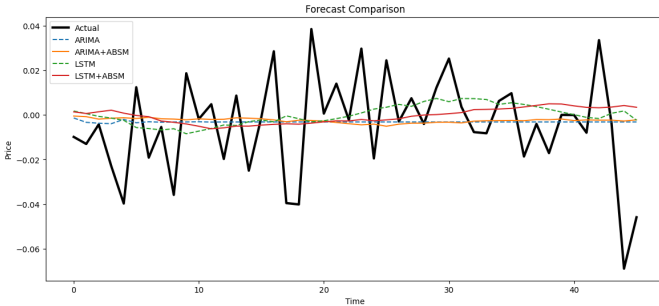


Fig. 3. Forecast comparison across models

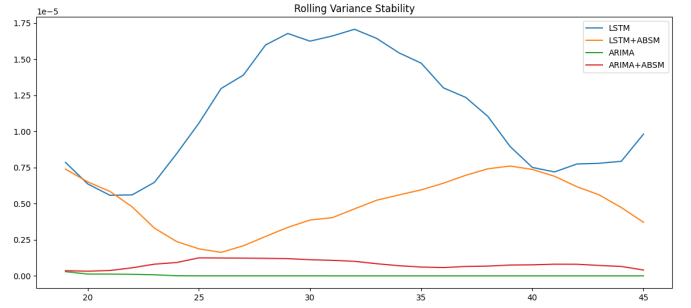


Fig. 4. Rolling variance showing ABSM stabilization effect

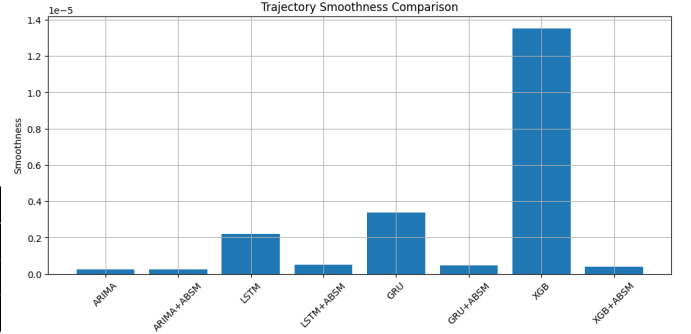


Fig. 5. Trajectory smoothness comparison

F. StockGro Portfolio Execution

The optimized portfolio generated from the forecasting and stabilization framework was executed on the StockGro virtual trading platform during the official capstone trading window.

Portfolio allocation emphasized relatively stable banking-sector assets alongside diversified market exposure through RELIANCE and TCS-based forecasting analysis.

TABLE IV
INITIAL NOTEBOOK-BASED PORTFOLIO ALLOCATION STRATEGY

Stock	Allocation (INR)
HDFCBANK	250000
ICICIBANK	200000
SBIN	200000
TCS	150000
RELIANCE	200000
Total	1000000

TABLE V
ADAPTIVE PORTFOLIO ALLOCATION EXECUTED DURING LIVE TRADING

Stock	Approx. Final Allocation (INR)
HDFCBANK	293500.62
ICICIBANK	277607.40
SBIN	242955.48
RELIANCE	175337.50
TCS	0

The adaptive allocation strategy deviated from the original notebook-generated allocation in response to observed live-market volatility, directional momentum, and short-term stability behavior during the execution window.

Particular emphasis was shifted toward relatively stable banking-sector assets such as HDFCBANK and ICICIBANK, while exposure to TCS and RELIANCE was reduced due to persistent downward trajectories and weaker short-term directional consistency.

This demonstrates the integration of model-driven allocation with practical real-time portfolio adaptation under dynamic market conditions.

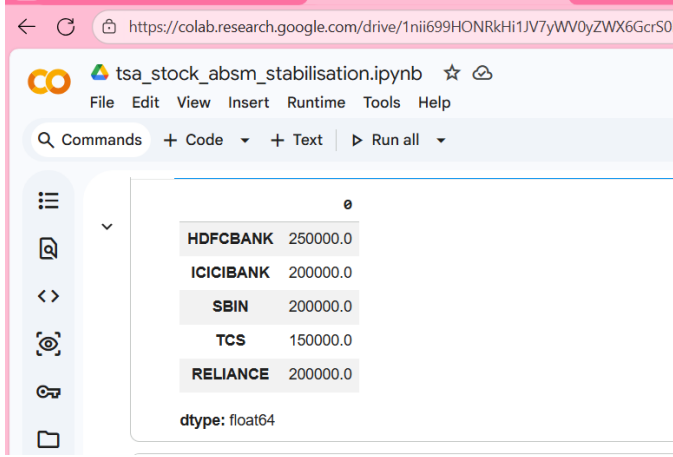


Fig. 6. Model-driven portfolio allocation strategy

The execution phase demonstrated how time-series forecasting outputs can be integrated into practical portfolio construction and dynamic allocation under real-time simulated market conditions.

Portfolio execution and observation were conducted during the official consecutive trading window of 13 May 2026 to 14 May 2026, in accordance with the capstone participation guidelines.

G. Key Findings

- ABSM consistently reduces trajectory variance across nonlinear models.
- LSTM and GRU benefit most in directional stability.
- ARIMA shows limited sensitivity to ABSM transformation.
- XGBoost demonstrates limited statistically significant distributional transformation under ABSM stabilization.
- Portfolio behavior is influenced more by volatility than raw error metrics.

VI. LIMITATIONS

The study has several limitations. First, the forecasting horizon is relatively short, which may not fully capture long-term market regimes. Second, the ABSM framework introduces stochastic perturbation, which may reduce interpretability in certain cases. Third, external

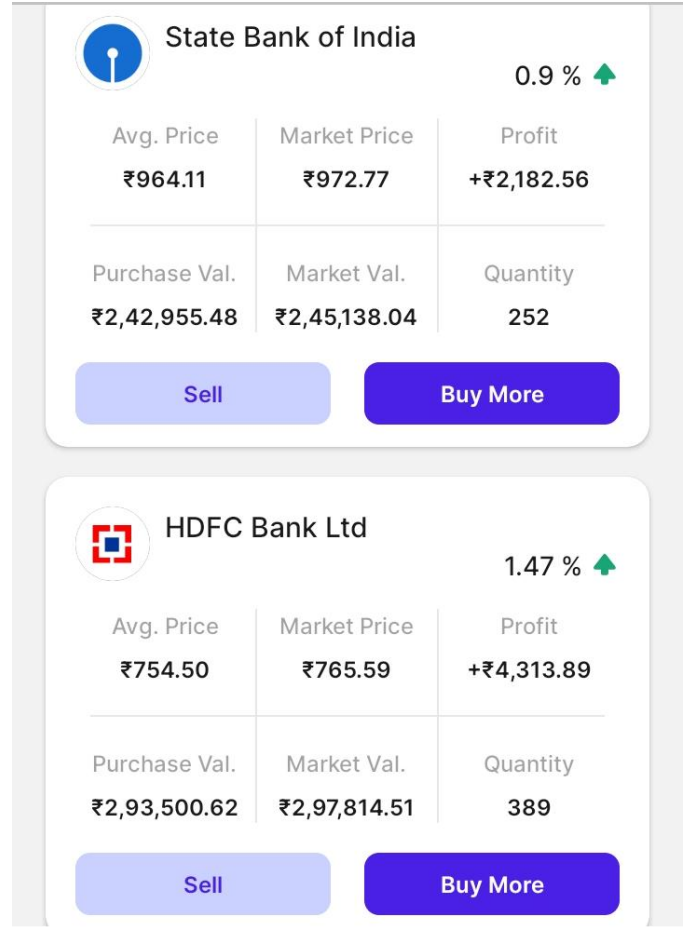


Fig. 7. Final StockGro portfolio allocation execution

macroeconomic variables and news sentiment were not incorporated, limiting real-world robustness. Finally, real trading costs, slippage, and liquidity constraints were not modeled in the simulation.

VII. CONCLUSION

This study demonstrates that ABSM acts as a post-prediction stabilization framework that improves temporal smoothness and reduces forecast volatility across financial time-series models. While classical models like ARIMA show limited sensitivity, nonlinear architectures such as LSTM, GRU, and XGBoost benefit significantly in terms of trajectory stability and directional consistency. Overall, ABSM is more effective in volatility reduction than direct error minimization, making it suitable for noisy financial environments and portfolio-level forecasting systems.

REPRODUCIBILITY

All experiments, preprocessing steps, forecasting pipelines, and ABSM stabilization implementations are reproducible.

Source code and experimental notebooks are available at:

[Github Repository](#)

Supplementary Google Colab implementation:

[Notebook](#)

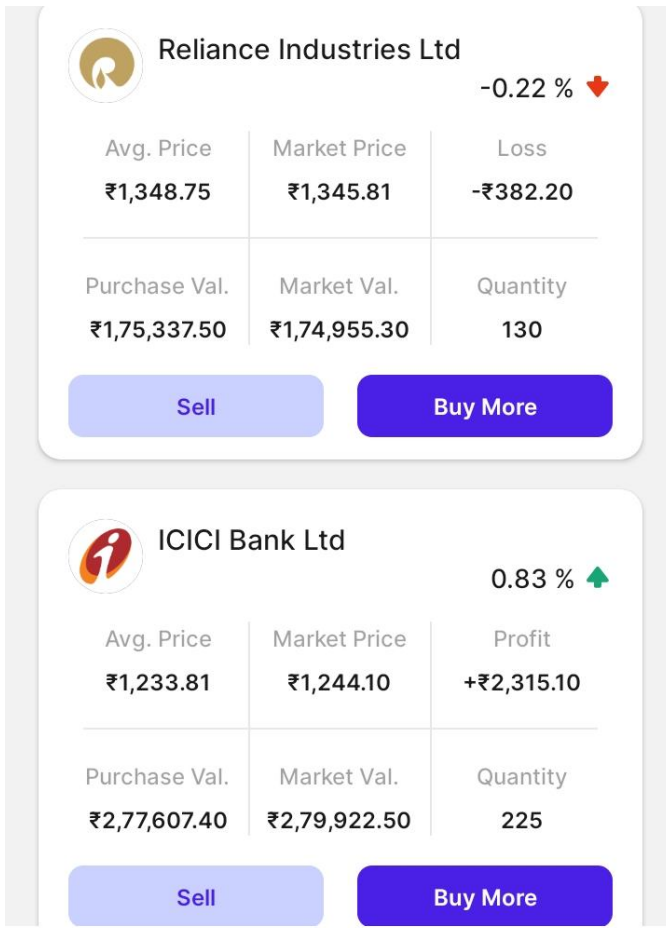


Fig. 8. Final portfolio allocation during live observation window

REFERENCES

- [1] G. E. P. Box and G. M. Jenkins, *Time Series Analysis: Forecasting and Control*, Holden-Day, 1970.
- [2] R. F. Engle, "Autoregressive Conditional Heteroskedasticity with Estimates of the Variance of United Kingdom Inflation," *Econometrica*, 1982.
- [3] S. Hochreiter and J. Schmidhuber, "Long Short-Term Memory," *Neural Computation*, 1997.
- [4] K. Cho et al., "Learning Phrase Representations using RNN Encoder-Decoder for Statistical Machine Translation," *EMNLP*, 2014.
- [5] P. Ghosh, "Adaptive Stochastic Contraction in Time-Varying Systems: A Unified Framework for Persistent Noise Dynamics," Zenodo, Mar. 2026. doi: 10.5281/zenodo.19100571. [Online]. Available: <https://doi.org/10.5281/zenodo.19100571>
- [6] P. Ghosh, "Adaptive Balancing Scaling Model (ABSM) for Single Computational Node," Figshare, Oct. 2025. doi: 10.6084/m9.figshare.30278944. [Online]. Available: <https://doi.org/10.6084/m9.figshare.30278944>
- [7] P. Ghosh, "Adaptive Balancing Scaling Model (ABSM): A Fixed-Point Framework for Stable Computational Transformation," Zenodo, Mar. 2026. doi: 10.5281/zenodo.19312138. [Online]. Available: <https://doi.org/10.5281/zenodo.19312138>
- [8] P. Ghosh, "Adaptive Post-Prediction Stabilization for Forecasting Models via Stochastic Contraction and ABSM Scaling," Zenodo, Apr. 2026. doi: 10.5281/zenodo.19896863 [Online]. Available: <https://doi.org/10.5281/zenodo.19896863>

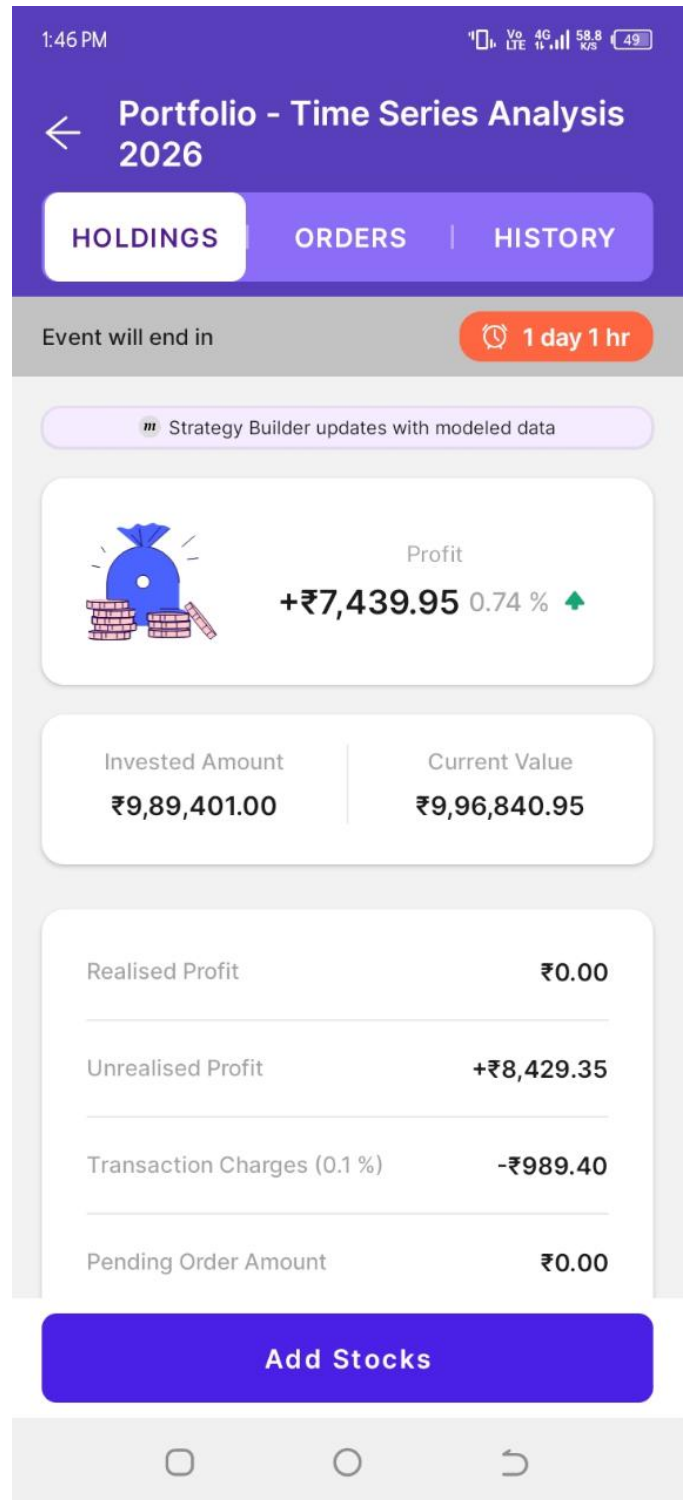


Fig. 9. StockGro live trading dashboard and portfolio performance